

Algorithmic Exam– L1 – S1 – 2024-2025

NO QUESTIONS WILL BE ANSWERED DURING THE EXAMINATION.

Exercise 1 (4 pts) :

Consider the following formula to calculate:

$$\text{Sine}(X) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^p \frac{x^{2p+1}}{(2p+1)!}$$

Write a Pascal function which receives as input a value of X and displays, after calculation according to the previous formula, the sine of x without using either a power function or a factorial function.

Note: we will stop for a value of P also given as a parameter.

Exercise 2 (6 pts) :

Write a procedure that eliminates identical consecutive letters from a character string.

Example: if we give the string « ELLE A CASSE ACCIDENTELEMMENT SA TASSE » to the procedure, the string becomes:

« ELE A CASE ACIDENTELEMMENT SA TASE » (the number of repeating letters is not limited to two).

Exercise 3: (4 pts)

We call the standard deviation of a series of values the following formula:

$$\sqrt{\frac{1}{n} \left(\sum_{i=1}^n x_i^2 \right) - \bar{x}^2}$$

- 1 – Write the average function which calculates the average of the elements of a vector (it is noted in the formula $\bar{x}$ with a bar at the top) (2 pts)
- 2 – Write the Ecart_type function which calculates the standard deviation of a series of values represented in a table x. (x_i in the formula is the i^{th} cell of the table x). (2 pts)

Exercise 4: (6 pts)

We would like to calculate the number of occurrences of each alphabetical letter in a character string. To do this, we use the “frequency” type defined by the user as follows:

Type

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frequency = record
  lettre : char ;
  Nbre : integer ;
End ;
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Where “letter” represents a letter in the string and “Nbre” is the number of times it appears.

- 1 - Write the operation which consists of filling in the “frequency” table. (3 pts)
- 2 – Write the function that returns the letter of the string with the greatest frequency (3 pts)